Surface Code on IQM Emerald: Circuit Verification Results

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Summary

44 of 49 tests pass; 5 fail. All failures are `test_roundtrip_matches_direct_stim[1-5]`, which stop because the decoder functions `decode_events` / `decode_ler` are not in the repository yet. No assertion about the circuit failed. With a temporary PyMatching stand-in decoder, the round-trip comparison agrees to within 1 standard error at every round count (section 5).

Experiment: distance-3 rotated surface code, Z-basis memory, no mid-circuit reset (Emerald cannot reset mid-circuit), 9 data + 8 ancilla = 17 physical qubits, 1 logical qubit, rounds 1 to 5. All numbers are from noiseless or simulated runs; nothing here ran on hardware. Simulated noise is uniform: p = 0.003 for gate depolarization, reset flips, measurement flips and per-round data depolarization.

1. What was built

Function	Returns / does
make_stim_circuit(distance=3, rounds=1, basis='Z', no_reset=True, noise=None)	Stim's rotated_memory_z circuit; with no_reset=True every MR becomes M and detectors are rewritten (to_noreset).
build_pair(distance, rounds, noise)	(hw, twin): noisy twin for simulation / decoder model; hw = twin.without_noise(), so both are the same circuit.
to_qiskit_hw(hw)	(qc, stim_to_dense, meas_order): drops the initial R (qubits start in |0>), converts to Qiskit; classical bit i = Stim measurement i.
noreset_tools.to_noreset(circuit)	No-reset rewrite: each referenced measurement j becomes {j, prev(j)}, so bulk detectors compare an ancilla two rounds apart. Stand-in for a team file that was missing; tested against stim_to_qrisp._analyse().

2. Test results (live run)

Command: `python -m pytest verification -q` → `5 failed, 44 passed in 4.28s`

File	Test	Result	Note
test_circuit.py	test_noiseless_detectors_and_observable_silent[1]	PASSED	
test_circuit.py	test_noiseless_detectors_and_observable_silent[2]	PASSED	
test_circuit.py	test_noiseless_detectors_and_observable_silent[3]	PASSED	
test_circuit.py	test_noiseless_detectors_and_observable_silent[4]	PASSED	
test_circuit.py	test_noiseless_detectors_and_observable_silent[5]	PASSED	
test_circuit.py	test_counts[1]	PASSED	
test_circuit.py	test_counts[2]	PASSED	
test_circuit.py	test_counts[3]	PASSED	
test_circuit.py	test_counts[4]	PASSED	
test_circuit.py	test_counts[5]	PASSED	
test_circuit.py	test_circuit_level_distance[True-1]	PASSED	
test_circuit.py	test_circuit_level_distance[True-2]	PASSED	
test_circuit.py	test_circuit_level_distance[True-3]	PASSED	
test_circuit.py	test_circuit_level_distance[True-4]	PASSED	
test_circuit.py	test_circuit_level_distance[True-5]	PASSED	
test_circuit.py	test_circuit_level_distance[False-1]	PASSED	
test_circuit.py	test_circuit_level_distance[False-2]	PASSED	
test_circuit.py	test_circuit_level_distance[False-3]	PASSED	
test_circuit.py	test_circuit_level_distance[False-4]	PASSED	
test_circuit.py	test_circuit_level_distance[False-5]	PASSED	
test_circuit.py	test_naive_mr_to_m_is_non_deterministic[2]	PASSED	
test_circuit.py	test_naive_mr_to_m_is_non_deterministic[3]	PASSED	

File	Test	Result	Note
test_circuit.py	test_naive_mr_to_m_is_non_deterministic[4]	PASSED	
test_circuit.py	test_naive_mr_to_m_is_non_deterministic[5]	PASSED	
test_circuit.py	test_no_reset_rewrite_matches_stim_to_qrisp[1]	PASSED	
test_circuit.py	test_no_reset_rewrite_matches_stim_to_qrisp[2]	PASSED	
test_circuit.py	test_no_reset_rewrite_matches_stim_to_qrisp[3]	PASSED	
test_circuit.py	test_no_reset_rewrite_matches_stim_to_qrisp[4]	PASSED	
test_circuit.py	test_no_reset_rewrite_matches_stim_to_qrisp[5]	PASSED	
test_circuit.py	test_build_pair_twins_match[1]	PASSED	
test_circuit.py	test_build_pair_twins_match[2]	PASSED	
test_circuit.py	test_build_pair_twins_match[3]	PASSED	
test_circuit.py	test_build_pair_twins_match[4]	PASSED	
test_circuit.py	test_build_pair_twins_match[5]	PASSED	
test_circuit.py	test_to_qiskit_hw[1]	PASSED	
test_circuit.py	test_to_qiskit_hw[2]	PASSED	
test_circuit.py	test_to_qiskit_hw[3]	PASSED	
test_circuit.py	test_to_qiskit_hw[4]	PASSED	
test_circuit.py	test_to_qiskit_hw[5]	PASSED	
test_pipeline.py	test_noiseless_aer_all_silent[1]	PASSED	
test_pipeline.py	test_noiseless_aer_all_silent[2]	PASSED	
test_pipeline.py	test_noiseless_aer_all_silent[3]	PASSED	
test_pipeline.py	test_noiseless_aer_all_silent[4]	PASSED	
test_pipeline.py	test_noiseless_aer_all_silent[5]	PASSED	
test_pipeline.py	test_roundtrip_matches_direct_stim[1]	FAILED	Failed: decode_events() not found in any of ['decoder', 'decoding', 'decode', 'run_on_hardware', 'surface_code
test_pipeline.py	test_roundtrip_matches_direct_stim[2]	FAILED	Failed: decode_events() not found in any of ['decoder', 'decoding', 'decode', 'run_on_hardware', 'surface_code
test_pipeline.py	test_roundtrip_matches_direct_stim[3]	FAILED	Failed: decode_events() not found in any of ['decoder', 'decoding', 'decode', 'run_on_hardware', 'surface_code
test_pipeline.py	test_roundtrip_matches_direct_stim[4]	FAILED	Failed: decode_events() not found in any of ['decoder', 'decoding', 'decode', 'run_on_hardware', 'surface_code
test_pipeline.py	test_roundtrip_matches_direct_stim[5]	FAILED	Failed: decode_events() not found in any of ['decoder', 'decoding', 'decode', 'run_on_hardware', 'surface_code

3. Circuit structure, measured

Noiseless no-reset circuit, 2,000 detector samples per round count. The two distance columns are the shortest error that flips the logical without firing any detector (circuit-level distance), for the no-reset and the with-reset circuit. 'MR->M, no rewrite' is the negative control.

r	Dets (8r)	Meas (8r+9)	Qubits used	num_ qubits	Dets fired	Obs flips	Dist no-reset	Dist reset	MR->M, no rewrite	Same as Qrisp
1	8	17	17	26	0	0	3	3	accepted (expected, r=1)	yes
2	16	25	17	26	0	0	3	3	rejected: non-det.	yes
3	24	33	17	26	0	0	3	3	rejected: non-det.	yes
4	32	41	17	26	0	0	3	3	rejected: non-det.	yes
5	40	49	17	26	0	0	3	3	rejected: non-det.	yes

At 1 round each ancilla is measured once, so the reset and no-reset circuits are identical and the naive swap is correctly accepted; the negative test runs for rounds 2 to 5.

4. Hardware-path checks (Qiskit / Aer)

Noiseless Qiskit circuit from to_qiskit_hw, 1,000 shots on Aer (stabilizer method), converted with counts_to_measurement_array and extract_syndromes. 'Varying raw columns' counts measurement bits that are not constant across shots: X-check outcomes and data qubits are random, so any bit-order or measurement-order bug would make detectors fire. Zero detectors firing despite this randomness is the evidence.

Rounds	hw = twin (coords + gates)	Twin error mechanisms	Qiskit qubits	Classical bits	Reset ops	Detectors fired	Observable flips	Varying raw columns
1	yes	23	17	17	0	0	0	13
2	yes	107	17	25	0	0	0	13
3	yes	219	17	33	0	0	0	17
4	yes	331	17	41	0	0	0	17
5	yes	443	17	49	0	0	0	21

Qrisp path

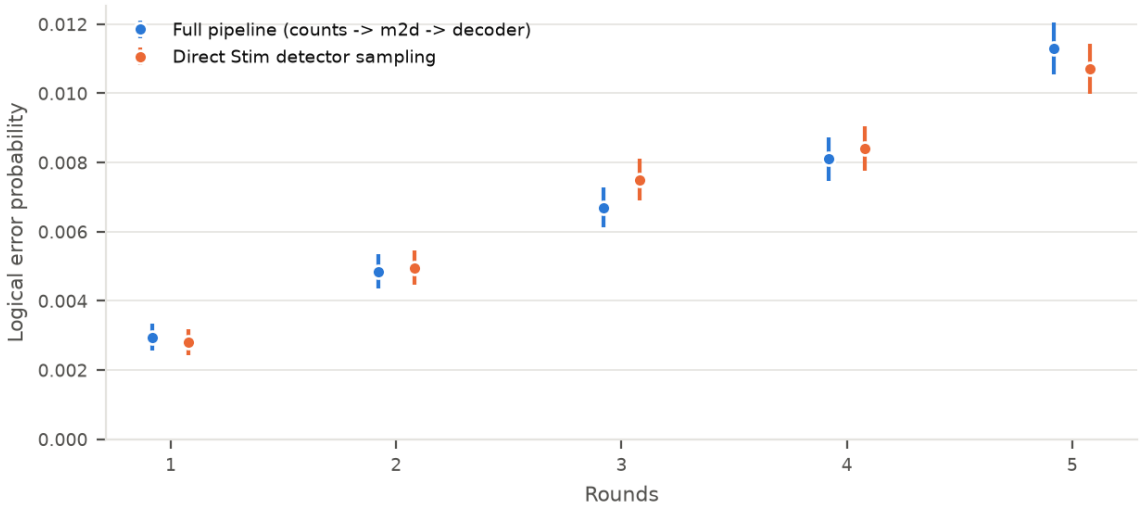
The same hw circuit through stim_to_detector_experiment(hw) and DetectorExperiment.to_stim(), 2,000 noiseless samples:

Rounds	Detectors	Detectors fired	Observable flips
1	8	0	0
2	16	0	0
3	24	0	0
4	32	0	0
5	40	0	0

5. Round trip: full pipeline vs. direct Stim

20,000 shots of the noisy twin, formatted as a Qiskit counts dictionary (rightmost character = classical bit 0), then counts_to_measurement_array → extract_syndromes → decoder, compared with decoding Stim's own detector samples. Pass criterion: |difference| <= 3 combined standard errors. **The decoder here is a temporary PyMatching stand-in, not C's decoder**; these rows show the test logic works, not the final pipeline.

Rounds	P, full pipeline	P, direct Stim	|diff| / sigma	Within 3 sigma
1	0.0029 ± 0.0004	0.0028 ± 0.0004	0.28	yes
2	0.0049 ± 0.0005	0.0050 ± 0.0005	0.14	yes
3	0.0067 ± 0.0006	0.0075 ± 0.0006	0.95	yes
4	0.0081 ± 0.0006	0.0084 ± 0.0006	0.33	yes
5	0.0113 ± 0.0007	0.0107 ± 0.0007	0.58	yes



Logical error probability after r rounds, d=3, p=0.003, 20,000 shots each; bars are 1 standard error. Stand-in decoder.

6. Open items

• **Decoder (C):** decode_events / decode_ler not in the repo. The round-trip test assumes decode_events(det_events, obs_flips, circuit) and decode_ler(circuit, shots), each returning P or (P, stderr); C should confirm or adjust.

• **noreset_tools.py:** written as a stand-in because the referenced team file was missing. If a team version exists, swap it in and rerun the tests.

• **Noise model:** in the no-reset twin, Stim's 'after reset' flip now acts on a live ancilla, a rough stand-in for idle decay. Hardware decoding should use calibration-based noise.

• **Not yet done:** X-basis memory and the CZ-native schedule (stretch goals), hardware runs, the logical error rate fit.